

Jasivan Alex SIVAKUMAR

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QUALIFICATIONS

- Sep 21 - Dec 25 **PhD Computer Science** (Supervised by [Dr. Nafise Sadat Moosavi](#))
- ▷ Natural Language Processing Group, University of Sheffield, United Kingdom
 - ▷ Research title: *Numerical Reasoning for General-Purpose Language Models*
 - ▷ Focus: AI, ML, DL, NLU, NLG, LLM, Reasoning, Representation Learning, Tokenisation
- Sep 20 - Sep 21 **MA Computational Linguistics** - Distinction
- ▷ University of Wolverhampton, United Kingdom
- Oct 12 - Jul 16 **Bachelor (MMath) Mathematics** - First Class
- ▷ University of Warwick, United Kingdom

PUBLICATIONS

- Jan 25 (COLING) **Sivakumar, J., & Moosavi, N. (2025).** [How to Leverage Digit Embeddings to Represent Numbers?](#) International Conference on Computational Linguistics.
- ▷ Designed a mathematically informed number representation by aggregating digit ones.
 - ▷ Evaluated the alignment of novel number representations with numerical proximity.
 - ▷ Explored incorporating new number representations at encoding and in loss function.
- Jul 23 (ACL) **Sivakumar, J., & Moosavi, N. (2023).** [FERMAT: An Alternative to Accuracy for Numerical Reasoning](#). Association for Computational Linguistics.
- ▷ Designed a mathematically informative multi-view test sets for numerical reasoning.
 - ▷ Published a method for automatic data-augmentation of worded arithmetic problem.
 - ▷ Explored data-leakage to justify improvement of finetuned BART and FLAN type LLMs.
- Dec 21 (IALP-IEEE) **Sivakumar, J., et. al. (2021).** [A GRU-based pipeline approach for word-sentence segmentation and punctuation restoration in English](#). International Conference on Asian Language Processing, Institute of Electrical and Electronics Engineers.
- ▷ Trained deep learning classification models using PyTorch for punctuation retrieval.
 - ▷ Generated synthetic training and testing from a big data source of punctuated text.
 - ▷ Investigated automatic evaluations metrics against human perception.

PROFESSIONAL ACTIVITIES

- Jul 24 - Oct 24 **Applied Scientist Intern** - Alexa Int., Amazon (Bellevue, United States)
- ▷ Created benchmark using AWS tools with human-in-the-loop approach for relevancy.
 - ▷ Optimised prompts of foundational models (Claude, Llama and Mistral) for given tasks.
 - ▷ Presented documentation to cross-functional partners to explain benchmarking need.
 - ▷ Collaborated with software engineers to review code and deploy updates to production.
- Jan 24 - Jun 24 **Consultancy** - Student Marketing / Elevate (Sheffield, United Kingdom)
- ▷ Researched and explained a RAG solution to automate high-volume query response.
 - ▷ Delivered a Topic Modelling solution to analyse large and complex document sources.
 - ▷ Advised clients on cost, efficiency, scalability, stability and technical skills required.

SKILLS

Programming

- ▷ Python (PyTorch, Huggingface, AWS, scikit-learn, NLTK)
- ▷ MATLAB
- ▷ Bash (HPC)
- ▷ Version control (git/github)

Languages

- ▷ French (Native)
- ▷ Tamil (Native)
- ▷ English (Native)
- ▷ Spanish (Fluent)

RESEARCH ACTIVITIES

Sep 21 - Present	NLP for Endangered Language Revitalisation in Colombia ▷ Collected and digitised text using OCR for Palenquero, an endangered language. ▷ Deploying low-resource language NLP research to develop pedagogical resource.
Nov 21 - Jun 22	Speech Processing and NLP to improve Oral History Search functionality ▷ Explored existing NER systems to identify domain specific terms using data annotation. ▷ Automated annotated transcriptions to improve search and retrieval of video frames.
Sep 20 - Sep 21	Semantic Sentence Embeddings for Natural Language Inference ▷ Trained BERT embeddings using siamese network contrastive loss over NLI tasks. ▷ Inductively biased embeddings with semantic parse trees over Graph Neural Network.
Sep 15 - May 16	A Mathematical Approach at Analysing the Influence of English on French ▷ Generated a diachronic dataset of French news articles to observe language evolution. ▷ Modelled adoption of English lexica into French with stochastic/differential equations.

ACADEMIC ACTIVITIES

Jan 24 - Present	Reviewer - ARR/ACL/EMNLP/NAACL/LREC/COLING ▷ Tracks: NLP and LLM Applications; Resources and Evaluation; Reasoning, Question Answering, and Sentence-level Semantics; Lexicon and Semantics, and Social Media Processing; Less-Resourced/Endangered/Less-studied Languages; Bio lay summarisation.
Nov 24	Invited talk - Chatbot for legal advice ▷ Presented a high-level functioning of LLMs in Chatbot to undergraduate law students discussing risk and benefits highlighting ethical and legal considerations.
Jan 22 - Dec 24	Human Annotator ▷ Contributed in data annotation for: Psychology (Eye-tracking from reading), Human-Robot Interaction (Perception of robot feature), Text (Hate Speech detections, Misinformation), Speech (Intelligibility), Vision (Naming Images, Bias from Visual Posters).
Nov 23 - Jun 24	Supervisor - final year student research project in number representation ▷ Weekly supervision for number decoding project, helped with latest deep learning techniques both theoretically and practically including transformers.
Oct 23 - Jun 24	Reading Group Co-organiser - Natural Language Processing ▷ Organised weekly reading group by securing a presenter and relevant paper, presented and facilitated discussion, arranged lightning conference paper sessions for Neurips and EMNLP
Jun 23	Invited Talk - 3 rd Speech and Language Technology CDT Conference ▷ Presented FERMAT research in main talk event.
Nov 22 - Feb 23	Examiner - Text Processing and Professional Issues ▷ Assessed student code and marked reports, provided feedback on assignments.
Jul 22	12th Lisbon Machine Learning Summer School ▷ Attended lectures on linear classifiers, seq2seq models, neural networks and talks from industry and academia on novel research, networked with other attendees.
Jan 22 - Jun 22	Teaching Assistant - Foundations of Computer Science ▷ Taught fundamental mathematics: probability, linear algebra and number theory.
Jan 21 - May 21	Tutorial - Machine Learning for Postgraduate Computational Linguists ▷ Explained mathematics using linguistics examples for kNN, decision trees, Naive Bayes, linear regressions, regularisation, beam search, EM algorithm, gradient descent, entropy.